BTS–II(R)&I(S)–05.16–0769 Reg. No.

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B.Tech. Degree II Semester Regular and I Semester Supplementary Examination May 2016

IT/CS/EC/CE/EE/ME/SE 1101 A / 1201 B CALCULUS

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Solve $\frac{dy}{dx} = e^{2x-y} + x^3 e^{-y}$.

(b) Solve $\frac{d^2y}{dx^2} + 25y = \cos 4x$.

(c) Solve $x^2 \frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = e^x$.

(d) Test the convergence of the series $\sum_{n=0}^{\infty}\left(\frac{1}{2^n}\right)$.

(e) Explain the Cauchy's root test and give one example.

(f) Obtain the Maclaurin series expansion of $f(x) = \sin x$.

(g) Find $\frac{\partial^3 u}{\partial x\, \partial y\, \partial z}$, *if* $u = e^{x^2+y^2+z^2}$.

(h) Find the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ of $u = x \sin y;\ v = y \sin x$.

(i) Find the area bounded by the parabolic arc $\sqrt{x} + \sqrt{y} = \sqrt{c}$, $c > o$ and the coordinate axes.

(j) Evaluate

$$\int_o^a \int_o^{\sqrt{a^2-y^2}} (x^2 + y^2)\, dx\, dy.$$

PART B

$(4 \times 10 = 40)$

II. (a) Solve $\frac{dy}{dx} = \frac{y-x}{y-x+2}$.

(b) The slope at any points (x,y) of a curve is $1 + \frac{y}{x}$. If the curve passes through $(1,1)$ find the equation of the curve.

(c) Solve $4y^2 p^2 + 2pxy(3+1) + 3x^3 = 0$, where $p = \frac{dy}{dx}$.

OR

(P.T.O.)

III. (a) Solve $\frac{dy}{dx}+2y=e^{x}(3\sin 2x+2\cos 2x)$.

(b) Solve $4x^{2}\frac{d^{2}y}{dx^{2}}-4x\frac{dy}{dx}+3y=0$.

(c) Solve the simultaneous equations

$$\frac{dy}{dt}+5y-2x=t$$

$$\frac{dx}{dt}+2y+x=0$$

IV. (a) Explain Raabe's test for testing the convergence of an infinite series and using it test the convergence of the series $\sum_{n=1}^{\infty}\left[\frac{4.7...(3n+1)}{1.2...n}\right]x^{n}$.

(b) Test for the uniform convergence of the series

$$\frac{\sin x}{1^{2}}+\frac{\sin 2x}{2^{2}}+\frac{\sin 3x}{3^{2}}+\frac{\sin 4x}{4^{2}}+....$$

OR

V. (a) Discuss the convergence of the series

$$1+\frac{x}{2}+\frac{x^{2}}{5}+....+\frac{x^{n}}{n^{2}+1}+.....$$

(b) Obtain power series solution in powers of x for the following differential equation: $\left(x^{2}-1\right)\frac{d^{2}y}{dx^{2}}+3x\frac{dy}{dx}+xy=0,\ y(0)=4$ *and* $y'(0)=6$.

VI (a) If $u=x^{3}y^{2}\sin^{-1}(y/x)$ show that

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=5u.$$

(b) Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube.

OR

VII. (a) If $V=u^{2}v$ and $u=e^{x^{2}-y^{2}}, v=\sin(xy^{2})$ find $\frac{dv}{dx}$ and $\frac{dv}{dy}$.

(b) Given $u=2axy,\ v=a(x^{2}-y^{2})$ where $x=r\cos\theta,\ \ y=r\sin\theta$, find $\frac{\partial(u,v)}{\partial(r,\theta)}$.

VIII. (a) Find the total area between the cubic $y=2x^{3}-3x^{2}-12x$, x-axis and its maximum and minimum ordinates.

(b) Evaluate $\iint_{R}(x+y)^{2}\,dxdy$, where R is the parallelogram in the xy-plane with vertices $(1,0), (3,1), (2,2), (0,1)$ using the transformation $u=x+y$ and $v=x-2y$.

OR

IX. (a) What is the volume generated by revolving the area enclosed by the loop of the curve $y^{4}=x(4-x)$ about x-axis?

(b) Find the mass, centroid of the tetrahedron bounded by the coordinate planes and the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$. Given that the substance has constant density ρ.

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